▣ 28 Confounding

Using the customer survey data, we constructed a regression model with spend as the response and with age and sex as predictors.

We found the regression coefficients for both to be statistically significant, indicating that there were two distinct effects.

1. Females tend to spend more than males
2. Older people tend to spend more.

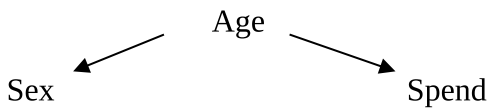
Suppose we found that spend was statistically significant, but sex was not. How would we interpret this?

Such a finding would suggest that there was no difference between mean spend by males and females and that we only observed such a difference because females tended to be older and older people spend more.

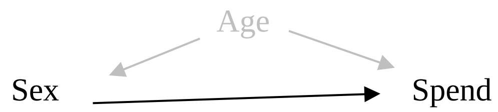
We could demonstrate this using regression if we have recorded both the age and gender of customers.

But suppose we didn't record their age and compared mean spend across the levels of gender (using a t test), we might very well establish that there was a difference, which might be misleading.

i.e. It may be that the age is the factor that causes the change in spending level but we might erroneously be attributing the cause to gender



True causation relationship?



Spurious causation relationship

We would say that age <u>confounds</u> the sex-spend relationship

Confounding can be a major problem in attempting to establish causation as the confounding variable may not even be collected,

For example, in the problem here perhaps it's `income` and not age that is the true causal factor of greater spends. (I.e. people's incomes tend to increase with age and those with higher incomes will tend to spend more)

In some cases, the confounding variable can be identified with a little reflection.

Example
An amateur gardener devises measures of both slug activity and plant growth in his garden, observes that they are positively correlated and so reasons that were he to increase the slug activity he might expect to see more rapid plant growth.

Spurious causation: Slug activity → Rapid plant growth

True causation

Example
An educator observes that the palm span of children is positively correlated with their reading ability.

Spurious causation: Palm span → Reading ability

True causation

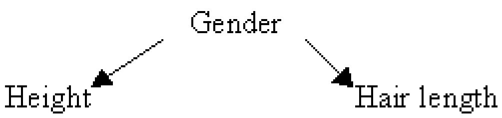
Maturity

Physical growth   Intellectual development

Example
It is widely observed that shorter people tend to have longer hair. Is there a psychological explanation?

No! Gender is a hidden confounder.

Women tend to be shorter than men (anatomical reason)
And women tend to wear their hair longer (cultural reason)



Correct causation relationship

Spurious causation relationship

Example
Data collected over a number of years showed a positive correlation between the number of televisions sold and the number of house fires. Is it reasonable to suspect that televisions are responsible for the fires?

Spurious causation: TV → House fires

True causation

Population
growth

More TVs sold   More house fires

Example
Statistics collected for a number of rural towns show a correlation between the number of public street lights and the burglary rate. Are street lights serving to assist the criminal classes??

Spurious causation: Street lights → Burglaries

True causation

Town size

More lights   More burglaries

N.B. Correlation does not imply causation

This is a caution that researchers, particularly in the social sciences, or any observational studies, must heed.

It is less of a problem (but still needs to be considered) in experimental studies.

Why is it more of a problem in the former?

Example

A teacher develops a novel method for national school children to enhance their numeracy skills, which she suggests might be better than the traditional method used.
To test this theory, two groups are recruited - one exposed to the traditional method (control group) and one with the new (treatment group).
A suitable test to measure the numeracy skill is administered and the two groups compared.

A critical consideration in the design of this investigation is the manner in which participants are assigned to the control and treatment group.

One approach is to let the students self-select – i.e. choose the group they wish to be in. An alternative approach for the investigator might be to randomly allocate each child to either the control or treatment group.

In the case of the former, the investigator does not interfere with the study or direct students to join any particular group – they simply <u>observe</u> all of the variables in the study.

In the case of the latter, the investigator does not merely observe, they specify the value of the "group" variable for each child – i.e. they <u>experiment</u>.

A potential problem with the observational is that there may be a hidden confounder.

Perhaps children who have greater mathematical skills may be more drawn to a novel approach while those with lesser skills may opt for the familiar.

And thus, a finding that the treatment group scored higher in the test may be explained by that group having greater ability to being with, and not that the treatment is better.

The same problem does not arise with the experimental study as allocation to groups is done randomly.

Of course, it often is not possible to do such random allocations. Perhaps all children in one class are allocated to the control group while those in another are allocated to the treatment group.

And of course, the classes may not have the same "baseline" ability.

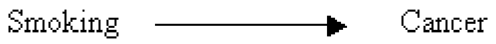
If we had a measure of this ability (perhaps a mark from an earlier, common examination, we could use this to statistically control for this confounder.

I.e. regress Test score on group (control v treatment) and ability.

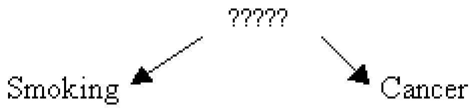
We could seek a finding of a difference between the control and treatment group on test score controlling for ability.

Example

What about smoking as a carcinogenic? Is there statistical “proof”?



Widely accepted as being the case



Can we rule out the possibility of a confounder?

The statistical evidence for smoking being a cause of cancer is that they are correlated.

To be confident that there is no hidden confounder we would need to devise a clinical trial with participants allocated at random to either a smoking group or a non-smoking group and then compare the incidence of cancer in the two groups,

But for ethical reasons, such a trial will not be conducted.

Multiple regression

A statistical method that is useful for dealing with confounding is multiple regression. But we do not consider this method in this module.